

Mathematical Working of Linear Regression

1. Linear Model Equation

Simple Linear Regression model:

$$y_{\text{hat}} = w \cdot x + b$$

Where:

y_{hat} = predicted value

x = input feature

w = weight (slope)

b = bias (intercept)

2. Residual (Error)

Residual is the difference between actual and predicted value.

$$\text{error}_i = y_i - y_{\text{hat}_i}$$

Where:

y_i = actual value

y_{hat_i} = predicted value

3. Mean Squared Error Cost Function

The cost function used in Linear Regression:

$$J(w, b) = (1/n) * \sum (y_i - y_{\text{hat}_i})^2$$

Substituting prediction equation:

$$J(w, b) = (1/n) * \sum (y_i - (w \cdot x_i + b))^2$$

4. Partial Derivative with respect to w

Cost function:

$$J(w, b) = (1/n) \sum (y_i - (w \cdot x_i + b))^2$$

Derivative:

$$\partial J / \partial w = (-2/n) \sum x_i (y_i - (w * x_i + b))$$

5. Partial Derivative with respect to b

Derivative of cost with respect to bias:

$$\partial J / \partial b = (-2/n) \sum (y_i - (w * x_i + b))$$

6. Gradient Descent Update Rules

Weights and bias are updated iteratively:

$$w = w - \alpha * (\partial J / \partial w)$$

$$b = b - \alpha * (\partial J / \partial b)$$

Where:

α = learning rate

7. Normal Equation (Closed Form Solution)

For multiple linear regression we can compute weights directly.

$$y = XW$$

Optimal weights:

$$W = (X^T X)^{-1} X^T y$$

Where:

X = feature matrix

X^T = transpose of X

$(X^T X)^{-1}$ = inverse matrix